

Complements of Operational Research

Exercise sheet 4 — Replacement theory

The exercises marked **Homework** are to be solved at home and sent by email to the lecturer, by the deadline given in class; the solution is shared afterwards. The others are from previous tests and exams. Unless stated otherwise, replacement is by an identical piece of equipment.

Exercise 1 — Garment-making machine

Homework (exam, 10/07/2018)

Consider the following table of a replacement problem for a garment-making machine.

Year	1	2	3	4	5	6	7	8
Running cost	200	500	800	1200	1800	2500	3200	4000

Given that the purchase price of the machine was 12 200 m.u. and the scrap value is 200 m.u., when should it be replaced?

Exercise 2 — Replacement with discounting (I)

Homework

A machine costs 60 000 m.u.; the discount rate is 10% and the scrap value is 0. The running costs are:

Year	1	2	3	4	5	6	7	8	9	10
Cost	10000	10000	10000	10000	10000	13000	16000	19000	22000	25000

When should the machine be replaced by an identical one?

Exercise 3 — Replacement with discounting (II)

Homework

Same statement as the previous exercise, with a machine cost of 500 000 m.u. (discount rate 10%, scrap value 0) and the running costs:

Year	1	2	3	4	5	6	7	8	9	10
Cost	60000	60000	60000	60000	60000	80000	100000	120000	140000	160000

When should the machine be replaced by an identical one?

Exercise 4 — Pump maintenance in a chemical plant

Homework

A chemical plant has 1000 centrifugal pumps, each essential for continuous operation. **Preventive maintenance** can be done at any time, at 2.5 m.u. per pump, and restores the pump to its initial condition. If a pump breaks down, **corrective maintenance** costs 2.5 m.u. per pump plus 7.5 m.u. for the unscheduled stoppage. The (cumulative) failure probability as a function of the time since the last maintenance is:

Months since the last maintenance	1	2	3	4	≥ 5
$P(t)$	0.10	0.25	0.50	0.80	1

- What is the optimal frequency of preventive maintenance?
- Compute the expected total annual cost of the optimal strategy.

Exercise 5 — Warehouse forklift

Homework (new)

A warehouse buys electric forklifts for 25 000 €. The running costs and the resale value at the end of each year are:

Year	1	2	3	4	5	6	7
Running cost (€)	2000	2500	3200	4200	5500	7200	9300
Resale (€)	18000	14000	11000	8500	6500	5000	4000

- Without considering the time value of money, after how many years should the forklift be replaced?
- And with a discount rate of 8%? (Cost of year i at the start of the year; resale at the end of year n .)
- Explain why the answers are different.

Exercise 6 — Machine with resale value

exam, 11/07/2019

Consider the following table of a machine replacement problem.

Year	1	2	3	4	5
Resale value	42000	28000	20400	14400	9650
Maintenance cost	4000	4290	5000	5700	6800
Consumption	14000	16000	18000	21000	25000

Given that the purchase price of the machine was 60 000, when should it be replaced?

Exercise 7 — Two machines

exam, 26/07/2019

Machine A costs 45 000 m.u. and its operating costs are 1000 m.u. in the first year and increase by 10 000 m.u. in each of the following years. Machine B costs 50 000 m.u. and its operating costs are 2000 m.u. in the first year and increase by 4000 m.u. in each of the following years. At the moment we have machine A in its first year of operation.

- Should machine A be replaced by machine B? Justify.
- If the answer to the previous part is yes, when should that replacement take place?

Exercise 8 — Metalworking machine with discounting

exam, 27/06/2019

Consider the following table of a replacement problem for a metalworking machine.

Year	1	2	3	4	5	6	7	8	9	10
Consumption	100	100	100	200	300	500	600	700	900	1000

Given that the purchase price of the machine was 1000, that the company expects a 10% return on investment (discount rate) and that the scrap value is 0, when should it be replaced?

Exercise 9 — Type AA safety valves

1st test, 13/04/2018

A refinery has 2000 type AA safety valves in its facilities. The individual replacement cost of each valve is 6 m.u., but if all the valves are replaced (replacing the whole group), the replacement cost is 2 m.u. per valve. The (cumulative) failure probabilities are given in the following table:

t (month)	1	2	3	4	5
$P(t)$	0.10	0.20	0.40	0.65	1

- What is the individual replacement cost per month?
- Determine the total cost and the right time for group replacement.
- Which of the two options is more advantageous? Justify.

Exercise 10 — Type AA safety valves

1st test, 08/04/2019

A factory has 10000 type AA safety valves in its facilities. The individual replacement cost of each valve is 50 m.u., but if all the valves are replaced (replacing the whole group), the replacement cost is 15 m.u. per valve. The (cumulative) failure probabilities are given in the following table:

t (year)	1	2	3	4	5
$P(t)$	0.10	0.35	0.70	0.90	1

- What is the individual replacement cost per year?
- Determine the total cost and the right time for group replacement.
- Which of the two options is more advantageous? Justify.

Exercise 11 — Light bulbs

exam, 05/07/2021

An engineering company has 1100 light bulbs in its facilities. The individual replacement cost of each bulb is 6 m.u., but if all the bulbs are replaced (replacing the whole group), the replacement cost is 1.20 m.u. per bulb. The (cumulative) failure probabilities are given in the following table:

t (month)	1	2	3	4	5
$P(t)$	0.10	0.25	0.45	0.70	1

- What is the mean life of a bulb?
- What is the individual replacement cost per month?
- What is the group replacement cost per month?
- What should the “optimal policy” to follow be?

Exercise 12 — Type AA centrifugal pumps

1st test, 03/04/2024

A chemical company has 500 type AA centrifugal pumps in its facilities. The individual replacement cost of each pump is 75 m.u., but if all the pumps are replaced (replacing the whole group), the replacement cost is 10 m.u. per pump. The (cumulative) failure probabilities are given in the following table:

t (month)	1	2	3	4	5
$P(t)$	0.10	0.25	0.50	0.75	1

- What is the individual replacement cost per month?
- Determine the total cost and the right time for group replacement.
- Compute the expected total annual cost of the optimal maintenance strategy.

Exercise 13 — Identical electric motors

1st test, 27/03/2025

A metalworking company has 300 identical electric motors in its facilities. The individual replacement cost of each motor is 120 m.u., but if all the motors are replaced (replacing the whole group), the replacement cost is 30 m.u. per motor. The (cumulative) failure probabilities are given in the following table:

t (month)	1	2	3	4	5
$P(t)$	0.15	0.35	0.60	0.85	1

- Compute the average monthly cost of individual replacement of the motors.
- Determine the total cost and the ideal month for group replacement.
- Determine the expected total annual cost of the optimal maintenance strategy.